could be written on the theoretical criteria for this, the practical rules are much simpler. Use as many samples as you *think* are necessary. After finding the frequency response, go back and repeat the procedure using twice as many samples. If the two frequency responses are adequately similar, you can be assured that the truncation of the impulse response hasn't fooled you in some way.

Cascade and Parallel Stages

Sophisticated recursive filters are usually designed in stages to simplify the tedious algebra of the z-domain. Figure 33-4 illustrates the two common ways that individual stages can be arranged: cascaded stages and parallel stages with added outputs. For example, a low-pass and high-pass stage can be cascaded to form a *band-pass* filter. Likewise, a parallel combination of low-pass and high-pass stages can form a *band-reject* filter. We will call the two stages being combined *system 1* and *system 2*, with their recursion coefficients being called: a_0, a_1, a_2, b_1, b_2 and A_0, A_1, A_2, B_1, B_2, respectively. Our goal is to combine these stages (in cascade or parallel) into a single recursive filter, which we will call *system 3*, with recursion coefficients given by: $a_0, a_1, a_2, a_3, a_4, b_1, b_2, b_3, b_4$.

As you recall from previous chapters, the frequency responses of systems in a cascade are combined by *multiplication*. Also, the frequency responses of systems in parallel are combined by *addition*. These same rules are followed by the z-domain transfer functions. This allows recursive systems to be combined by moving the problem into the z-domain, performing the required multiplication or addition, and then returning to the recursion coefficients of the final system.

As an example of this method, we will work out the algebra for combining two biquad stages in a cascade. The transfer function of each stage is found by writing Eq. 33-3 using the appropriate recursion coefficients. The transfer function of the entire system, $H[z]$, is then found by multiplying the transfer functions of the two stage:

$$H[z] = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 - b_1 z^{-1} - b_2 z^{-2}} \times \frac{A_0 + A_1 z^{-1} + A_2 z^{-2}}{1 - B_1 z^{-1} - B_2 z^{-2}}$$

Multiplying out the polynomials and collecting like terms:

$$H[z] = \frac{a_0 A_0 + (a_0 A_1 + a_1 A_0) z^{-1} + (a_0 A_2 + a_1 A_1 + a_2 A_0) z^{-2} + (a_1 A_2 + a_2 A_1) z^{-3} + (a_2 A_2) z^{-4}}{1 - (b_1 + B_1) z^{-1} - (b_2 + B_2 - b_1 B_1) z^{-2} - (-b_1 B_2 - b_2 B_1) z^{-3} - (-b_2 B_2) z^{-4}}$$